SG No. 141 D_{4h}^{19} $I4_1/amd$ [tetragonal]

* plus set: $+[0,\ 0,\ 0]$, $+\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$
* Wyckoff site: `4a`, site symmetry: `-4m2`

Table 1: Wyckoff bond: `4a@4a`, bond symmetry: `-4m2`

No.	vector	center	mapping
1	$[0,\ 0,\ Z]$	$\left[0,\ \frac{3}{4},\ \frac{1}{8}\right]$	`[1,2,-7,-8,-11,-12,13,14]`
2	$[0,\ 0,\ Z]$	$\left[\frac{1}{2},\ \frac{3}{4},\ \frac{3}{8}\right]$	`[3,4,-5,-6,-9,-10,15,16]`

Table 2: Wyckoff bond: `8b@4a`, bond symmetry: `2mm.`

No.	vector	center	mapping
1	$[X,\ 0,\ 0]$	$\left[0,\ \frac{3}{4},\ \frac{1}{8}\right]$	`[1,-2,13,-14]`
2	$[0,\ X,\ 0]$	$\left[\frac{1}{2},\ \frac{3}{4},\ \frac{3}{8}\right]$	`[3,-4,-15,16]`
3	$[-X,\ 0,\ 0]$	$\left[\frac{1}{2},\ \frac{3}{4},\ \frac{3}{8}\right]$	`[5,-6,9,-10]`
4	$[0,\ X,\ 0]$	$\left[0,\ \frac{3}{4},\ \frac{1}{8}\right]$	`[7,-8,-11,12]`

Table 3: Wyckoff bond: `8c@4a`, bond symmetry: `2.22`

No.	vector	center	mapping
1	$[X,\ X,\ 0]$	$\left[0,\ \frac{3}{4},\ \frac{1}{8}\right]$	`[1,-2,7,-8]`
2	$[-X,\ X,\ 0]$	$\left[\frac{1}{2},\ \frac{3}{4},\ \frac{3}{8}\right]$	`[3,-4,5,-6]`
3	$[-X,\ -X,\ 0]$	$\left[\frac{1}{2},\ \frac{3}{4},\ \frac{3}{8}\right]$	`[9,-10,15,-16]`
4	$[X,\ -X,\ 0]$	$\left[0,\ \frac{3}{4},\ \frac{1}{8}\right]$	`[11,-12,13,-14]`

Table 4: Wyckoff bond: `16d@4a`, bond symmetry: `.m.`

No.	vector	center	mapping
1	$[X,\ 0,\ Z]$	$\left[0,\ \frac{3}{4},\ \frac{1}{8}\right]$	`[1,13]`
2	$[-X,\ 0,\ Z]$	$\left[0,\ \frac{3}{4},\ \frac{1}{8}\right]$	`[2,14]`
3	$[0,\ X,\ Z]$	$\left[\frac{1}{2},\ \frac{3}{4},\ \frac{3}{8}\right]$	`[3,16]`
4	$[0,\ -X,\ Z]$	$\left[\frac{1}{2},\ \frac{3}{4},\ \frac{3}{8}\right]$	`[4,15]`
5	$[-X,\ 0,\ -Z]$	$\left[\frac{1}{2},\ \frac{3}{4},\ \frac{3}{8}\right]$	`[5,9]`
6	$[X,\ 0,\ -Z]$	$\left[\frac{1}{2},\ \frac{3}{4},\ \frac{3}{8}\right]$	`[6,10]`
7	$[0,\ X,\ -Z]$	$\left[0,\ \frac{3}{4},\ \frac{1}{8}\right]$	`[7,12]`
8	$[0,\ -X,\ -Z]$	$\left[0,\ \frac{3}{4},\ \frac{1}{8}\right]$	`[8,11]`

Table 5: Wyckoff bond: `16e@4a`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X,\ X,\ Z]$	$\left[0,\ \frac{3}{4},\ \frac{1}{8}\right]$	`[1,-8]`
2	$[-X,\ -X,\ Z]$	$\left[0,\ \frac{3}{4},\ \frac{1}{8}\right]$	`[2,-7]`
3	$[-X,\ X,\ Z]$	$\left[\frac{1}{2},\ \frac{3}{4},\ \frac{3}{8}\right]$	`[3,-6]`
4	$[X,\ -X,\ Z]$	$\left[\frac{1}{2},\ \frac{3}{4},\ \frac{3}{8}\right]$	`[4,-5]`

continued ...

Table 5

No.	vector	center	mapping
5	$[-X, -X, -Z]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{3}{8}\right]$	`[9,-16]`
6	$[X, X, -Z]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{3}{8}\right]$	`[10,-15]`
7	$[X, -X, -Z]$	$\left[0, \frac{3}{4}, \frac{1}{8}\right]$	`[11,-14]`
8	$[-X, X, -Z]$	$\left[0, \frac{3}{4}, \frac{1}{8}\right]$	`[12,-13]`

Table 6: Wyckoff bond: `16f@4a`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[0, \frac{3}{4}, \frac{1}{8}\right]$	`[1,-2]`
2	$[-Y, X, 0]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{3}{8}\right]$	`[3,-4]`
3	$[-X, Y, 0]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{3}{8}\right]$	`[5,-6]`
4	$[Y, X, 0]$	$\left[0, \frac{3}{4}, \frac{1}{8}\right]$	`[7,-8]`
5	$[-X, -Y, 0]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{3}{8}\right]$	`[9,-10]`
6	$[Y, -X, 0]$	$\left[0, \frac{3}{4}, \frac{1}{8}\right]$	`[11,-12]`
7	$[X, -Y, 0]$	$\left[0, \frac{3}{4}, \frac{1}{8}\right]$	`[13,-14]`
8	$[-Y, -X, 0]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{3}{8}\right]$	`[15,-16]`

Table 7: Wyckoff bond: `32g@4a`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[0, \frac{3}{4}, \frac{1}{8}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[0, \frac{3}{4}, \frac{1}{8}\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{3}{8}\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{3}{8}\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{3}{8}\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{3}{8}\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[0, \frac{3}{4}, \frac{1}{8}\right]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[0, \frac{3}{4}, \frac{1}{8}\right]$	`[8]`
9	$[-X, -Y, -Z]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{3}{8}\right]$	`[9]`
10	$[X, Y, -Z]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{3}{8}\right]$	`[10]`
11	$[Y, -X, -Z]$	$\left[0, \frac{3}{4}, \frac{1}{8}\right]$	`[11]`
12	$[-Y, X, -Z]$	$\left[0, \frac{3}{4}, \frac{1}{8}\right]$	`[12]`
13	$[X, -Y, Z]$	$\left[0, \frac{3}{4}, \frac{1}{8}\right]$	`[13]`
14	$[-X, Y, Z]$	$\left[0, \frac{3}{4}, \frac{1}{8}\right]$	`[14]`
15	$[-Y, -X, Z]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{3}{8}\right]$	`[15]`
16	$[Y, X, Z]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{3}{8}\right]$	`[16]`

* Wyckoff site: `4b`, site symmetry: `-4m2`

Table 8: Wyckoff bond: `4a@4b`, bond symmetry: `-4m2`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[0, \frac{1}{4}, \frac{3}{8}\right]$	`[1,2,-7,-8,-11,-12,13,14]`
2	$[0, 0, Z]$	$\left[0, \frac{3}{4}, \frac{5}{8}\right]$	`[3,4,-5,-6,-9,-10,15,16]`

Table 9: Wyckoff bond: `8b@4b`, bond symmetry: `2mm.`

No.	vector	center	mapping
1	$[X, 0, 0]$	$[0, \frac{1}{4}, \frac{3}{8}]$	`[1,-2,13,-14]`
2	$[0, X, 0]$	$[0, \frac{3}{4}, \frac{5}{8}]$	`[3,-4,-15,16]`
3	$[-X, 0, 0]$	$[0, \frac{3}{4}, \frac{5}{8}]$	`[5,-6,9,-10]`
4	$[0, X, 0]$	$[0, \frac{1}{4}, \frac{3}{8}]$	`[7,-8,-11,12]`

Table 10: Wyckoff bond: `8c@4b`, bond symmetry: `2.22`

No.	vector	center	mapping
1	$[X, X, 0]$	$[0, \frac{1}{4}, \frac{3}{8}]$	`[1,-2,7,-8]`
2	$[-X, X, 0]$	$[0, \frac{3}{4}, \frac{5}{8}]$	`[3,-4,5,-6]`
3	$[-X, -X, 0]$	$[0, \frac{3}{4}, \frac{5}{8}]$	`[9,-10,15,-16]`
4	$[X, -X, 0]$	$[0, \frac{1}{4}, \frac{3}{8}]$	`[11,-12,13,-14]`

Table 11: Wyckoff bond: `16d@4b`, bond symmetry: `.m.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$[0, \frac{1}{4}, \frac{3}{8}]$	`[1,13]`
2	$[-X, 0, Z]$	$[0, \frac{1}{4}, \frac{3}{8}]$	`[2,14]`
3	$[0, X, Z]$	$[0, \frac{3}{4}, \frac{5}{8}]$	`[3,16]`
4	$[0, -X, Z]$	$[0, \frac{3}{4}, \frac{5}{8}]$	`[4,15]`
5	$[-X, 0, -Z]$	$[0, \frac{3}{4}, \frac{5}{8}]$	`[5,9]`
6	$[X, 0, -Z]$	$[0, \frac{3}{4}, \frac{5}{8}]$	`[6,10]`
7	$[0, X, -Z]$	$[0, \frac{1}{4}, \frac{3}{8}]$	`[7,12]`
8	$[0, -X, -Z]$	$[0, \frac{1}{4}, \frac{3}{8}]$	`[8,11]`

Table 12: Wyckoff bond: `16e@4b`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, X, Z]$	$[0, \frac{1}{4}, \frac{3}{8}]$	`[1,-8]`
2	$[-X, -X, Z]$	$[0, \frac{1}{4}, \frac{3}{8}]$	`[2,-7]`
3	$[-X, X, Z]$	$[0, \frac{3}{4}, \frac{5}{8}]$	`[3,-6]`
4	$[X, -X, Z]$	$[0, \frac{3}{4}, \frac{5}{8}]$	`[4,-5]`
5	$[-X, -X, -Z]$	$[0, \frac{3}{4}, \frac{5}{8}]$	`[9,-16]`
6	$[X, X, -Z]$	$[0, \frac{3}{4}, \frac{5}{8}]$	`[10,-15]`
7	$[X, -X, -Z]$	$[0, \frac{1}{4}, \frac{3}{8}]$	`[11,-14]`
8	$[-X, X, -Z]$	$[0, \frac{1}{4}, \frac{3}{8}]$	`[12,-13]`

Table 13: Wyckoff bond: `16f@4b`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$[0, \frac{1}{4}, \frac{3}{8}]$	`[1,-2]`
2	$[-Y, X, 0]$	$[0, \frac{3}{4}, \frac{5}{8}]$	`[3,-4]`

continued ...

Table 13

No.	vector	center	mapping
3	$[-X, Y, 0]$	$[0, \frac{3}{4}, \frac{5}{8}]$	`[5,-6]`
4	$[Y, X, 0]$	$[0, \frac{1}{4}, \frac{3}{8}]$	`[7,-8]`
5	$[-X, -Y, 0]$	$[0, \frac{3}{4}, \frac{5}{8}]$	`[9,-10]`
6	$[Y, -X, 0]$	$[0, \frac{1}{4}, \frac{3}{8}]$	`[11,-12]`
7	$[X, -Y, 0]$	$[0, \frac{1}{4}, \frac{3}{8}]$	`[13,-14]`
8	$[-Y, -X, 0]$	$[0, \frac{3}{4}, \frac{5}{8}]$	`[15,-16]`

Table 14: Wyckoff bond: `32g@4b`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, \frac{1}{4}, \frac{3}{8}]$	`[1]`
2	$[-X, -Y, Z]$	$[0, \frac{1}{4}, \frac{3}{8}]$	`[2]`
3	$[-Y, X, Z]$	$[0, \frac{3}{4}, \frac{5}{8}]$	`[3]`
4	$[Y, -X, Z]$	$[0, \frac{3}{4}, \frac{5}{8}]$	`[4]`
5	$[-X, Y, -Z]$	$[0, \frac{3}{4}, \frac{5}{8}]$	`[5]`
6	$[X, -Y, -Z]$	$[0, \frac{3}{4}, \frac{5}{8}]$	`[6]`
7	$[Y, X, -Z]$	$[0, \frac{1}{4}, \frac{3}{8}]$	`[7]`
8	$[-Y, -X, -Z]$	$[0, \frac{1}{4}, \frac{3}{8}]$	`[8]`
9	$[-X, -Y, -Z]$	$[0, \frac{3}{4}, \frac{5}{8}]$	`[9]`
10	$[X, Y, -Z]$	$[0, \frac{3}{4}, \frac{5}{8}]$	`[10]`
11	$[Y, -X, -Z]$	$[0, \frac{1}{4}, \frac{3}{8}]$	`[11]`
12	$[-Y, X, -Z]$	$[0, \frac{1}{4}, \frac{3}{8}]$	`[12]`
13	$[X, -Y, Z]$	$[0, \frac{1}{4}, \frac{3}{8}]$	`[13]`
14	$[-X, Y, Z]$	$[0, \frac{1}{4}, \frac{3}{8}]$	`[14]`
15	$[-Y, -X, Z]$	$[0, \frac{3}{4}, \frac{5}{8}]$	`[15]`
16	$[Y, X, Z]$	$[0, \frac{3}{4}, \frac{5}{8}]$	`[16]`

* Wyckoff site: `8c`, site symmetry: `.2/m.`

Table 15: Wyckoff bond: `8a@8c`, bond symmetry: `.2/m.`

No.	vector	center	mapping
1	$[0, X, Z]$	$[0, 0, 0]$	`[1,-6,-9,14]`
2	$[0, -X, Z]$	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[2,-5,-10,13]`
3	$[-X, 0, Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[3,-7,-11,15]`
4	$[X, 0, Z]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	`[4,-8,-12,16]`

Table 16: Wyckoff bond: `8b@8c`, bond symmetry: `.2/m.`

No.	vector	center	mapping
1	$[X, 0, 0]$	$[0, 0, 0]$	`[1,6,-9,-14]`
2	$[-X, 0, 0]$	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[2,5,-10,-13]`
3	$[0, X, 0]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[3,7,-11,-15]`
4	$[0, -X, 0]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	`[4,8,-12,-16]`

Table 17: Wyckoff bond: `16c@8c`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, 0, 0]$	`[1,-9]`
2	$[-X, -Y, Z]$	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[2,-10]`
3	$[-Y, X, Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[3,-11]`
4	$[Y, -X, Z]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	`[4,-12]`
5	$[-X, Y, -Z]$	$[\frac{1}{2}, 0, \frac{1}{2}]$	`[5,-13]`
6	$[X, -Y, -Z]$	$[0, 0, 0]$	`[6,-14]`
7	$[Y, X, -Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}]$	`[7,-15]`
8	$[-Y, -X, -Z]$	$[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}]$	`[8,-16]`

* Wyckoff site: `8d`, site symmetry: `.2/m.`

Table 18: Wyckoff bond: `8a@8d`, bond symmetry: `.2/m.`

No.	vector	center	mapping
1	$[0, X, Z]$	$[0, 0, \frac{1}{2}]$	`[1,-6,-9,14]`
2	$[0, -X, Z]$	$[\frac{1}{2}, 0, 0]$	`[2,-5,-10,13]`
3	$[-X, 0, Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[3,-7,-11,15]`
4	$[X, 0, Z]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[4,-8,-12,16]`

Table 19: Wyckoff bond: `8b@8d`, bond symmetry: `.2/m.`

No.	vector	center	mapping
1	$[X, 0, 0]$	$[0, 0, \frac{1}{2}]$	`[1,6,-9,-14]`
2	$[-X, 0, 0]$	$[\frac{1}{2}, 0, 0]$	`[2,5,-10,-13]`
3	$[0, X, 0]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[3,7,-11,-15]`
4	$[0, -X, 0]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[4,8,-12,-16]`

Table 20: Wyckoff bond: `16c@8d`, bond symmetry: `-1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, 0, \frac{1}{2}]$	`[1,-9]`
2	$[-X, -Y, Z]$	$[\frac{1}{2}, 0, 0]$	`[2,-10]`
3	$[-Y, X, Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[3,-11]`
4	$[Y, -X, Z]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[4,-12]`
5	$[-X, Y, -Z]$	$[\frac{1}{2}, 0, 0]$	`[5,-13]`
6	$[X, -Y, -Z]$	$[0, 0, \frac{1}{2}]$	`[6,-14]`
7	$[Y, X, -Z]$	$[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}]$	`[7,-15]`
8	$[-Y, -X, -Z]$	$[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}]$	`[8,-16]`

* Wyckoff site: `8e`, site symmetry: `2mm.`

Table 21: Wyckoff bond: `8a@8e`, bond symmetry: `2mm.`

No.	vector	center	mapping
1	$[X, 0, 0]$	$\left[0, \frac{1}{4}, z\right]$	`[1,-2,13,-14]`
2	$[0, X, 0]$	$\left[0, \frac{3}{4}, z+\frac{1}{4}\right]$	`[3,-4,-15,16]`
3	$[-X, 0, 0]$	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{2}-z\right]$	`[5,-6,9,-10]`
4	$[0, X, 0]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}-z\right]$	`[7,-8,-11,12]`

Table 22: Wyckoff bond: `8b@8e`, bond symmetry: `2mm.`

No.	vector	center	mapping
1	$[0, 0, Z]$	$\left[0, \frac{1}{4}, z\right]$	`[1,2,13,14]`
2	$[0, 0, Z]$	$\left[0, \frac{3}{4}, z+\frac{1}{4}\right]$	`[3,4,15,16]`
3	$[0, 0, -Z]$	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{2}-z\right]$	`[5,6,9,10]`
4	$[0, 0, -Z]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}-z\right]$	`[7,8,11,12]`

Table 23: Wyckoff bond: `16c@8e`, bond symmetry: `.m.`

No.	vector	center	mapping
1	$[X, 0, Z]$	$\left[0, \frac{1}{4}, z\right]$	`[1,13]`
2	$[-X, 0, Z]$	$\left[0, \frac{1}{4}, z\right]$	`[2,14]`
3	$[0, X, Z]$	$\left[0, \frac{3}{4}, z+\frac{1}{4}\right]$	`[3,16]`
4	$[0, -X, Z]$	$\left[0, \frac{3}{4}, z+\frac{1}{4}\right]$	`[4,15]`
5	$[-X, 0, -Z]$	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{2}-z\right]$	`[5,9]`
6	$[X, 0, -Z]$	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{2}-z\right]$	`[6,10]`
7	$[0, X, -Z]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}-z\right]$	`[7,12]`
8	$[0, -X, -Z]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}-z\right]$	`[8,11]`

Table 24: Wyckoff bond: `16d@8e`, bond symmetry: `2..`

No.	vector	center	mapping
1	$[X, Y, 0]$	$\left[0, \frac{1}{4}, z\right]$	`[1,-2]`
2	$[-Y, X, 0]$	$\left[0, \frac{3}{4}, z+\frac{1}{4}\right]$	`[3,-4]`
3	$[-X, Y, 0]$	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{2}-z\right]$	`[5,-6]`
4	$[Y, X, 0]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}-z\right]$	`[7,-8]`
5	$[-X, -Y, 0]$	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{2}-z\right]$	`[9,-10]`
6	$[Y, -X, 0]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}-z\right]$	`[11,-12]`
7	$[X, -Y, 0]$	$\left[0, \frac{1}{4}, z\right]$	`[13,-14]`
8	$[-Y, -X, 0]$	$\left[0, \frac{3}{4}, z+\frac{1}{4}\right]$	`[15,-16]`

Table 25: Wyckoff bond: `32e@8e`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[0, \frac{1}{4}, z\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[0, \frac{1}{4}, z\right]$	`[2]`

continued ...

Table 25

No.	vector	center	mapping
3	$[-Y, X, Z]$	$\left[0, \frac{3}{4}, z+\frac{1}{4}\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[0, \frac{3}{4}, z+\frac{1}{4}\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{2}-z\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{2}-z\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}-z\right]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}-z\right]$	`[8]`
9	$[-X, -Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{2}-z\right]$	`[9]`
10	$[X, Y, -Z]$	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{2}-z\right]$	`[10]`
11	$[Y, -X, -Z]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}-z\right]$	`[11]`
12	$[-Y, X, -Z]$	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}-z\right]$	`[12]`
13	$[X, -Y, Z]$	$\left[0, \frac{1}{4}, z\right]$	`[13]`
14	$[-X, Y, Z]$	$\left[0, \frac{1}{4}, z\right]$	`[14]`
15	$[-Y, -X, Z]$	$\left[0, \frac{3}{4}, z+\frac{1}{4}\right]$	`[15]`
16	$[Y, X, Z]$	$\left[0, \frac{3}{4}, z+\frac{1}{4}\right]$	`[16]`

* Wyckoff site: `16f`, site symmetry: `.2.`

Table 26: Wyckoff bond: `16a@16f`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[0, X, Z]$	$[x, 0, 0]$	`[1,-6]`
2	$[0, -X, Z]$	$\left[\frac{1}{2}-x, 0, \frac{1}{2}\right]$	`[2,-5]`
3	$[-X, 0, Z]$	$\left[\frac{1}{4}, x+\frac{3}{4}, \frac{1}{4}\right]$	`[3,-7]`
4	$[X, 0, Z]$	$\left[\frac{1}{4}, \frac{1}{4}-x, \frac{3}{4}\right]$	`[4,-8]`
5	$[0, -X, -Z]$	$[-x, 0, 0]$	`[9,-14]`
6	$[0, X, -Z]$	$\left[x+\frac{1}{2}, 0, \frac{1}{2}\right]$	`[10,-13]`
7	$[X, 0, -Z]$	$\left[\frac{3}{4}, \frac{1}{4}-x, \frac{3}{4}\right]$	`[11,-15]`
8	$[-X, 0, -Z]$	$\left[\frac{3}{4}, x+\frac{3}{4}, \frac{1}{4}\right]$	`[12,-16]`

Table 27: Wyckoff bond: `16b@16f`, bond symmetry: `.2.`

No.	vector	center	mapping
1	$[X, 0, 0]$	$[x, 0, 0]$	`[1,6]`
2	$[-X, 0, 0]$	$\left[\frac{1}{2}-x, 0, \frac{1}{2}\right]$	`[2,5]`
3	$[0, X, 0]$	$\left[\frac{1}{4}, x+\frac{3}{4}, \frac{1}{4}\right]$	`[3,7]`
4	$[0, -X, 0]$	$\left[\frac{1}{4}, \frac{1}{4}-x, \frac{3}{4}\right]$	`[4,8]`
5	$[-X, 0, 0]$	$[-x, 0, 0]$	`[9,14]`
6	$[X, 0, 0]$	$\left[x+\frac{1}{2}, 0, \frac{1}{2}\right]$	`[10,13]`
7	$[0, -X, 0]$	$\left[\frac{3}{4}, \frac{1}{4}-x, \frac{3}{4}\right]$	`[11,15]`
8	$[0, X, 0]$	$\left[\frac{3}{4}, x+\frac{3}{4}, \frac{1}{4}\right]$	`[12,16]`

Table 28: Wyckoff bond: `32c@16f`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, 0, 0]$	`[1]`
2	$[-X, -Y, Z]$	$[\frac{1}{2} - x, 0, \frac{1}{2}]$	`[2]`
3	$[-Y, X, Z]$	$[\frac{1}{4}, x + \frac{3}{4}, \frac{1}{4}]$	`[3]`
4	$[Y, -X, Z]$	$[\frac{1}{4}, \frac{1}{4} - x, \frac{3}{4}]$	`[4]`
5	$[-X, Y, -Z]$	$[\frac{1}{2} - x, 0, \frac{1}{2}]$	`[5]`
6	$[X, -Y, -Z]$	$[x, 0, 0]$	`[6]`
7	$[Y, X, -Z]$	$[\frac{1}{4}, x + \frac{3}{4}, \frac{1}{4}]$	`[7]`
8	$[-Y, -X, -Z]$	$[\frac{1}{4}, \frac{1}{4} - x, \frac{3}{4}]$	`[8]`
9	$[-X, -Y, -Z]$	$[-x, 0, 0]$	`[9]`
10	$[X, Y, -Z]$	$[x + \frac{1}{2}, 0, \frac{1}{2}]$	`[10]`
11	$[Y, -X, -Z]$	$[\frac{3}{4}, \frac{1}{4} - x, \frac{3}{4}]$	`[11]`
12	$[-Y, X, -Z]$	$[\frac{3}{4}, x + \frac{3}{4}, \frac{1}{4}]$	`[12]`
13	$[X, -Y, Z]$	$[x + \frac{1}{2}, 0, \frac{1}{2}]$	`[13]`
14	$[-X, Y, Z]$	$[-x, 0, 0]$	`[14]`
15	$[-Y, -X, Z]$	$[\frac{3}{4}, \frac{1}{4} - x, \frac{3}{4}]$	`[15]`
16	$[Y, X, Z]$	$[\frac{3}{4}, x + \frac{3}{4}, \frac{1}{4}]$	`[16]`

* Wyckoff site: `16g`, site symmetry: `..2`

Table 29: Wyckoff bond: `16a@16g`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, -X, Z]$	$[x, x + \frac{1}{4}, \frac{7}{8}]$	`[1,-7]`
2	$[-X, X, Z]$	$[\frac{1}{2} - x, \frac{3}{4} - x, \frac{3}{8}]$	`[2,-8]`
3	$[X, X, Z]$	$[-x, x + \frac{3}{4}, \frac{1}{8}]$	`[3,-5]`
4	$[-X, -X, Z]$	$[x + \frac{1}{2}, \frac{1}{4} - x, \frac{5}{8}]$	`[4,-6]`
5	$[-X, X, -Z]$	$[-x, \frac{3}{4} - x, \frac{1}{8}]$	`[9,-15]`
6	$[X, -X, -Z]$	$[x + \frac{1}{2}, x + \frac{1}{4}, \frac{5}{8}]$	`[10,-16]`
7	$[-X, -X, -Z]$	$[x, \frac{1}{4} - x, \frac{7}{8}]$	`[11,-13]`
8	$[X, X, -Z]$	$[\frac{1}{2} - x, x + \frac{3}{4}, \frac{3}{8}]$	`[12,-14]`

Table 30: Wyckoff bond: `16b@16g`, bond symmetry: `..2`

No.	vector	center	mapping
1	$[X, X, 0]$	$[x, x + \frac{1}{4}, \frac{7}{8}]$	`[1,7]`
2	$[-X, -X, 0]$	$[\frac{1}{2} - x, \frac{3}{4} - x, \frac{3}{8}]$	`[2,8]`
3	$[-X, X, 0]$	$[-x, x + \frac{3}{4}, \frac{1}{8}]$	`[3,5]`
4	$[X, -X, 0]$	$[x + \frac{1}{2}, \frac{1}{4} - x, \frac{5}{8}]$	`[4,6]`
5	$[-X, -X, 0]$	$[-x, \frac{3}{4} - x, \frac{1}{8}]$	`[9,15]`
6	$[X, X, 0]$	$[x + \frac{1}{2}, x + \frac{1}{4}, \frac{5}{8}]$	`[10,16]`
7	$[X, -X, 0]$	$[x, \frac{1}{4} - x, \frac{7}{8}]$	`[11,13]`
8	$[-X, X, 0]$	$[\frac{1}{2} - x, x + \frac{3}{4}, \frac{3}{8}]$	`[12,14]`

Table 31: Wyckoff bond: `32c@16g`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$\left[x, x+\frac{1}{4}, \frac{7}{8}\right]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}-x, \frac{3}{4}-x, \frac{3}{8}\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[-x, x+\frac{3}{4}, \frac{1}{8}\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[x+\frac{1}{2}, \frac{1}{4}-x, \frac{5}{8}\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[-x, x+\frac{3}{4}, \frac{1}{8}\right]$	`[5]`
6	$[X, -Y, -Z]$	$\left[x+\frac{1}{2}, \frac{1}{4}-x, \frac{5}{8}\right]$	`[6]`
7	$[Y, X, -Z]$	$\left[x, x+\frac{1}{4}, \frac{7}{8}\right]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[\frac{1}{2}-x, \frac{3}{4}-x, \frac{3}{8}\right]$	`[8]`
9	$[-X, -Y, -Z]$	$\left[-x, \frac{3}{4}-x, \frac{1}{8}\right]$	`[9]`
10	$[X, Y, -Z]$	$\left[x+\frac{1}{2}, x+\frac{1}{4}, \frac{5}{8}\right]$	`[10]`
11	$[Y, -X, -Z]$	$\left[x, \frac{1}{4}-x, \frac{7}{8}\right]$	`[11]`
12	$[-Y, X, -Z]$	$\left[\frac{1}{2}-x, x+\frac{3}{4}, \frac{3}{8}\right]$	`[12]`
13	$[X, -Y, Z]$	$\left[x, \frac{1}{4}-x, \frac{7}{8}\right]$	`[13]`
14	$[-X, Y, Z]$	$\left[\frac{1}{2}-x, x+\frac{3}{4}, \frac{3}{8}\right]$	`[14]`
15	$[-Y, -X, Z]$	$\left[-x, \frac{3}{4}-x, \frac{1}{8}\right]$	`[15]`
16	$[Y, X, Z]$	$\left[x+\frac{1}{2}, x+\frac{1}{4}, \frac{5}{8}\right]$	`[16]`

* Wyckoff site: `16h`, site symmetry: `.m.`

Table 32: Wyckoff bond: `16a@16h`, bond symmetry: `.m.`

No.	vector	center	mapping
1	$[0, X, Z]$	$[0, y, z]$	`[1,14]`
2	$[0, -X, Z]$	$\left[\frac{1}{2}, -y, z+\frac{1}{2}\right]$	`[2,13]`
3	$[-X, 0, Z]$	$\left[\frac{1}{4}-y, \frac{3}{4}, z+\frac{1}{4}\right]$	`[3,15]`
4	$[X, 0, Z]$	$\left[y+\frac{1}{4}, \frac{1}{4}, z+\frac{3}{4}\right]$	`[4,16]`
5	$[0, X, -Z]$	$\left[\frac{1}{2}, y, \frac{1}{2}-z\right]$	`[5,10]`
6	$[0, -X, -Z]$	$[0, -y, -z]$	`[6,9]`
7	$[X, 0, -Z]$	$\left[y+\frac{1}{4}, \frac{3}{4}, \frac{1}{4}-z\right]$	`[7,11]`
8	$[-X, 0, -Z]$	$\left[\frac{1}{4}-y, \frac{1}{4}, \frac{3}{4}-z\right]$	`[8,12]`

Table 33: Wyckoff bond: `16b@16h`, bond symmetry: `.m.`

No.	vector	center	mapping
1	$[X, 0, 0]$	$[0, y, z]$	`[1,-14]`
2	$[-X, 0, 0]$	$\left[\frac{1}{2}, -y, z+\frac{1}{2}\right]$	`[2,-13]`
3	$[0, X, 0]$	$\left[\frac{1}{4}-y, \frac{3}{4}, z+\frac{1}{4}\right]$	`[3,-15]`
4	$[0, -X, 0]$	$\left[y+\frac{1}{4}, \frac{1}{4}, z+\frac{3}{4}\right]$	`[4,-16]`
5	$[-X, 0, 0]$	$\left[\frac{1}{2}, y, \frac{1}{2}-z\right]$	`[5,-10]`
6	$[X, 0, 0]$	$[0, -y, -z]$	`[6,-9]`
7	$[0, X, 0]$	$\left[y+\frac{1}{4}, \frac{3}{4}, \frac{1}{4}-z\right]$	`[7,-11]`
8	$[0, -X, 0]$	$\left[\frac{1}{4}-y, \frac{1}{4}, \frac{3}{4}-z\right]$	`[8,-12]`

Table 34: Wyckoff bond: `32c@16h`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[0, y, z]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}, -y, z+\frac{1}{2}\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[\frac{1}{4}-y, \frac{3}{4}, z+\frac{1}{4}\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[y+\frac{1}{4}, \frac{1}{4}, z+\frac{3}{4}\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[\frac{1}{2}, y, \frac{1}{2}-z\right]$	`[5]`
6	$[X, -Y, -Z]$	$[0, -y, -z]$	`[6]`
7	$[Y, X, -Z]$	$\left[y+\frac{1}{4}, \frac{3}{4}, \frac{1}{4}-z\right]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[\frac{1}{4}-y, \frac{1}{4}, \frac{3}{4}-z\right]$	`[8]`
9	$[-X, -Y, -Z]$	$[0, -y, -z]$	`[9]`
10	$[X, Y, -Z]$	$\left[\frac{1}{2}, y, \frac{1}{2}-z\right]$	`[10]`
11	$[Y, -X, -Z]$	$\left[y+\frac{1}{4}, \frac{3}{4}, \frac{1}{4}-z\right]$	`[11]`
12	$[-Y, X, -Z]$	$\left[\frac{1}{4}-y, \frac{1}{4}, \frac{3}{4}-z\right]$	`[12]`
13	$[X, -Y, Z]$	$\left[\frac{1}{2}, -y, z+\frac{1}{2}\right]$	`[13]`
14	$[-X, Y, Z]$	$[0, y, z]$	`[14]`
15	$[-Y, -X, Z]$	$\left[\frac{1}{4}-y, \frac{3}{4}, z+\frac{1}{4}\right]$	`[15]`
16	$[Y, X, Z]$	$\left[y+\frac{1}{4}, \frac{1}{4}, z+\frac{3}{4}\right]$	`[16]`

* Wyckoff site: `32i`, site symmetry: `1`

Table 35: Wyckoff bond: `32a@32i`, bond symmetry: `1`

No.	vector	center	mapping
1	$[X, Y, Z]$	$[x, y, z]$	`[1]`
2	$[-X, -Y, Z]$	$\left[\frac{1}{2}-x, -y, z+\frac{1}{2}\right]$	`[2]`
3	$[-Y, X, Z]$	$\left[\frac{1}{4}-y, x+\frac{3}{4}, z+\frac{1}{4}\right]$	`[3]`
4	$[Y, -X, Z]$	$\left[y+\frac{1}{4}, \frac{1}{4}-x, z+\frac{3}{4}\right]$	`[4]`
5	$[-X, Y, -Z]$	$\left[\frac{1}{2}-x, y, \frac{1}{2}-z\right]$	`[5]`
6	$[X, -Y, -Z]$	$[x, -y, -z]$	`[6]`
7	$[Y, X, -Z]$	$\left[y+\frac{1}{4}, x+\frac{3}{4}, \frac{1}{4}-z\right]$	`[7]`
8	$[-Y, -X, -Z]$	$\left[\frac{1}{4}-y, \frac{1}{4}-x, \frac{3}{4}-z\right]$	`[8]`
9	$[-X, -Y, -Z]$	$[-x, -y, -z]$	`[9]`
10	$[X, Y, -Z]$	$\left[x+\frac{1}{2}, y, \frac{1}{2}-z\right]$	`[10]`
11	$[Y, -X, -Z]$	$\left[y+\frac{3}{4}, \frac{1}{4}-x, \frac{3}{4}-z\right]$	`[11]`
12	$[-Y, X, -Z]$	$\left[\frac{3}{4}-y, x+\frac{3}{4}, \frac{1}{4}-z\right]$	`[12]`
13	$[X, -Y, Z]$	$\left[x+\frac{1}{2}, -y, z+\frac{1}{2}\right]$	`[13]`
14	$[-X, Y, Z]$	$[-x, y, z]$	`[14]`
15	$[-Y, -X, Z]$	$\left[\frac{3}{4}-y, \frac{1}{4}-x, z+\frac{3}{4}\right]$	`[15]`
16	$[Y, X, Z]$	$\left[y+\frac{3}{4}, x+\frac{3}{4}, z+\frac{1}{4}\right]$	`[16]`